Capstone Project Proposal

Project Title

Release Risk Intelligence Platform: Predicting Software Release Risk Using GitHub Repository Analytics

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Program

Master of Data Analytics

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1. Introduction

Software organizations continuously release new versions of their products to deliver features, security updates, and bug fixes. However, software releases vary significantly in complexity and quality. Factors such as high code churn, unresolved issues, large numbers of pull requests, and frequent contributor changes can increase the likelihood of release instability and post-release defects.

Traditionally, release readiness decisions rely heavily on engineering judgment and experience. With the increasing availability of software repository data, organizations have an opportunity to adopt data-driven approaches for assessing release risk.

This project proposes the development of a Release Risk Intelligence Platform that leverages GitHub repository analytics to identify patterns associated with software release risk and provide predictive insights for release management.

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2. Problem Statement

Software repositories contain valuable information about development activity, including commits, pull requests, issues, contributors, and release history. Although these metrics may provide indicators of release quality, many organizations lack systematic methods for analyzing this information to assess release readiness.

As a result, release teams may deploy software without sufficient visibility into potential risks, increasing the likelihood of defects, service disruptions, and costly post-release fixes.

The proposed project seeks to determine whether software repository analytics can be used to predict software release risk and support better release management decisions.

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3. Research Question

Primary Research Question

Can GitHub repository metrics be used to predict the risk level of software releases?

Secondary Research Questions

1. Which repository metrics have the strongest influence on software release risk?
2. Does increased code churn contribute to higher release risk?
3. How do unresolved issues affect software release quality?
4. Can contributor activity serve as a reliable indicator of release readiness?
5. Which machine learning model provides the most accurate release risk classification?

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4. Research Hypotheses

Null Hypothesis (H_0)

GitHub repository metrics such as commits, pull requests, issues, contributor activity, and code churn do not have a statistically significant relationship with software release risk.

Alternative Hypothesis (H_1)

GitHub repository metrics such as commits, pull requests, issues, contributor activity, and code churn have a statistically significant relationship with software release risk and can be used to predict release risk levels.

Supporting Hypotheses

H_1a: Higher code churn is associated with higher software release risk.

H_1b: Releases with a larger number of unresolved issues are more likely to be classified as high risk.

H_1c: Longer issue resolution times increase software release risk.

H_1d: Increased pull request activity immediately before release increases release complexity and risk.

H_1e: Machine learning models can classify release risk significantly better than random prediction.

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5. Project Objectives

The objectives of this project are:

1. Collect release-related data from public GitHub repositories.
2. Extract repository metrics including commits, pull requests, issues, and contributor activity.
3. Perform exploratory data analysis to identify release risk patterns.
4. Develop a release risk scoring framework.
5. Build machine learning models for release risk classification.
6. Develop an interactive dashboard for release intelligence and monitoring.
7. Generate actionable recommendations to improve release management practices.

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6. Data Sources

The project will utilize publicly available data from GitHub repositories through the GitHub REST API.

Data Sources

- GitHub Releases
- GitHub Commits
- GitHub Pull Requests
- GitHub Issues
- Contributor Activity

Candidate Repositories

- entity["software","Apache Airflow","Workflow orchestration platform"]
- entity["software","Visual Studio Code","Microsoft code editor"]
- entity["software","Home Assistant","Open-source home automation platform"]

These repositories provide extensive release histories, active issue tracking, and rich development activity suitable for analytics and machine learning.

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7. Proposed Methodology

Phase 1: Data Collection

Data will be extracted using the GitHub REST API.

Collected data will include:

- Release versions
- Release dates
- Commit history
- Pull request activity
- Issue tracking information
- Contributor information

Phase 2: Data Preparation

Activities include:

- Data cleaning
- Missing value treatment
- Duplicate removal
- Feature engineering
- Release-level aggregation

Potential features include:

- Number of commits
- Number of pull requests
- Number of contributors
- Open issues
- Closed issues
- Average issue resolution time
- Release interval
- Code churn metrics

Phase 3: Exploratory Data Analysis

EDA will be performed to:

- Identify release trends
- Analyze contributor behavior
- Examine defect patterns
- Investigate relationships among release metrics

Phase 4: Predictive Modeling

Machine learning models will be developed to classify release risk into:

- Low Risk
- Medium Risk
- High Risk

Potential algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

Model performance will be evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC (if applicable)

Phase 5: Dashboard Development

An interactive dashboard will be developed using Power BI or Streamlit.

Dashboard components will include:

Executive Overview

- Total Releases
- High-Risk Releases
- Repository Activity Summary

Release Analytics

- Release Frequency Trends
- Commit Trends
- Contributor Trends

Risk Intelligence

- Release Risk Scores
- Feature Importance Analysis
- Risk Classification Results

Recommendations

- Suggested Release Actions
- Risk Mitigation Insights

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8. Expected Deliverables

1. GitHub release analytics dataset
2. Data preparation and feature engineering pipeline
3. Exploratory data analysis report
4. Release risk prediction model
5. Interactive dashboard
6. Final capstone report
7. Presentation and project demonstration

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9. Tools and Technologies

- Python
- Pandas
- NumPy
- Scikit-learn
- GitHub REST API
- SQL
- Power BI / Streamlit
- Databricks (optional)
- Git

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10. Expected Outcomes

The project is expected to demonstrate that software repository metrics can provide meaningful insights into software release quality and risk. The resulting Release Risk Intelligence Platform will support data-driven decision-making by identifying high-risk releases before deployment and providing recommendations for improving release readiness.

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11. Project Timeline

Week 1

Proposal approval and repository selection

Week 2–3

Data collection and integration

Week 4

Data cleaning and feature engineering

Week 5

Exploratory data analysis

Week 6

Machine learning model development

Week 7

Dashboard development

Week 8

Documentation, testing, presentation preparation, and final submission